Answer QUESTION 2 PARTS A and B on this page.

PART A

$r(\theta) = 2\sin^2\theta$

$\frac{dr}{d\theta} = 4\sin\theta\cos\theta$

$\frac{dr}{d\theta}$

$r'(\theta) = 4\sin\theta\cos\theta$

$r'(1.3) = 1.031$

$\cancel{r'(\theta) = 4\sin\theta}$

$\cancel{r'(1.3) = 3.854}$

PART B

$\frac{1}{2}\int_0^{\pi} (2\sin^2\theta)^2\, d\theta - \frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \left(\frac{1}{2}\right)^2 d\theta = \frac{5\pi}{12}$

$2\sin^2\theta = \frac{1}{2}$

$\sin^2\theta = \frac{1}{4}$

$\sin\theta = \frac{1}{2}$

$\frac{\pi}{6}, \frac{5\pi}{6}$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Answer QUESTION 2 PARTS C and D on this page.

PART C

$\cancel{4}\sin\theta\cos^2\theta - \cancel{2}\sin^3\theta = 0$

$2\sin\theta\cos^2\theta - \sin^3\theta = 0$

$2\sin\theta\cos^2\theta = \sin^3\theta$

0

$2\cos^2\theta = \sin^2\theta$

PART D

$\frac{d\theta}{dt} = 15$

$\frac{dr}{dt} = 2\sin\theta\cos\theta \frac{d\theta}{dt} 15$

15.465

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

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